

```
In [20]: import tensorflow as tf
        from tensorflow.keras.models import Sequential
        from tensorflow.keras.layers import Dense
        from tensorflow.keras.optimizers import Adam
        from tensorflow.keras.callbacks import EarlyStopping
        from keras_tuner import RandomSearch
        from keras_tuner.engine.hyperparameters import HyperParameters
        from sklearn.datasets import make_classification
        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import StandardScaler
```

```
In [22]: X,y=make_classification(n_samples=1000,n_features=20,n_classes=2,random_state=42)
```

```
In [23]: X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

```
In [24]: scaler=StandardScaler()
        X_train=scaler.fit_transform(X_train)
        X_test=scaler.transform(X_test)
```

```
In [32]: def build_model(hp):
        model=Sequential()
        model.add(Dense(units=hp.Int('units_1',min_value=16,max_value=128,step=16),activation='relu',input_dim=20))
        model.add(Dense(units=hp.Int('units_2',min_value=16,max_value=128,step=16),activation='relu'))
        model.add(Dense(1,activation='sigmoid'))
        model.compile(optimizer=Adam(learning_rate=hp.Float('learning_rate',min_value=0.0001,max_value=0.01,sampling='LOG')),
                      loss='binary_crossentropy',metrics=['accuracy'])
        return model
```

```
In [26]: tuner=RandomSearch(
        build_model,
        objective='val_accuracy',
        max_trials=10,
        executions_per_trial=2,
        directory='mydir',
        project_name='mlp_tuning'
    )
```

```
C:\Users\subha\anaconda3\envs\tf_env\lib\site-packages\keras\src\layers\core\dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.  
    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
In [29]: early_stopping=EarlyStopping(monitor='val_loss',patience=5,restore_best_weights=True)  
        tuner.search(X_train,y_train,validation_split=0.2,epochs=50,batch_size=32,callbacks=[early_stopping])
```

```
Trial 10 Complete [00h 00m 09s]  
val_accuracy: 0.875
```

```
Best val_accuracy So Far: 0.909375011920929  
Total elapsed time: 00h 02m 54s
```

```
In [30]: best_model=tuner.get_best_models(num_models=1)[0]  
        loss,accuracy=best_model.evaluate(X_test,y_test)  
        print(f"Test Accuracy after tuning: {accuracy*100:.2f}%")
```

```
C:\Users\subha\anaconda3\envs\tf_env\lib\site-packages\keras\src\saving\saving_lib.py:802: UserWarning: Skipping variable loading for optimizer 'adam', because it has 2 variables whereas the saved optimizer has 14 variables.  
    saveable.load_own_variables(weights_store.get(inner_path))
```

```
7/7 ————— 0s 14ms/step - accuracy: 0.8684 - loss: 0.3725  
Test Accuracy after tuning: 87.00%
```